TEXORA · LECTURE SHEET

Preparatory Machinery

Singeing, desizing, scouring, bleaching and mercerising machines decide the ceiling of everything downstream: no dyeing machine can repair a fabric that preparation has made uneven. This sheet maps the machines to the processes they serve, with the throughput arithmetic of a continuous range.

WP-3207 · ADVANCED MACHINERIES AND RELEVANT CALCULATIONS · TOPIC 6 · SPECIALIZATION — WET PROCESSING

1. The preparation sequence and its machines

Grey fabric carries fibre impurities (waxes, pectins, seed fragments), applied size (starch, PVA) and handling soils. Preparation removes them in a fixed order — you cannot scour effectively through a size film, nor bleach evenly over wax. Each stage exists as batch machines (for knits and short runs) and as continuous ranges (for woven volume production).

STAGE	PURPOSE	PRINCIPAL MACHINES
Singeing	Burn off protruding fibres for a smooth, clear surface	Gas singeing machine (burner rows over a cooled fabric path)
Desizing	Remove size film (enzyme, oxidative or hot-wash routes)	Pad–batch and pad–steam units, rot-steep tanks, J-box with saturator
Scouring	Remove waxes, pectins, oils; make the fabric absorbent	Kier (batch, cotton), J-box and roller steamer (continuous), soft-flow machines (knits)
Bleaching	Destroy natural colouring matter to a stable white	J-box peroxide ranges, pad–steam bleaching, hypochlorite units (legacy), knit bleach in soft-flow
Mercerising	Caustic treatment under tension for lustre, strength and dye affinity	Chain (pin/clip) merceriser and chainless merceriser, with caustic recovery
Heat setting	Stabilise thermoplastic fabrics before dyeing	Stenter (Topic 9) run as a pre-set range

2. The continuous preparation range

For woven cotton, the stages are linked into one line: singe → desize pad → J-box dwell → wash → scour saturator → steamer → wash → bleach saturator → steamer → wash → dry. The fabric moves in rope or open width at a set speed; each chemical is applied by a **saturator** (padder) and given reaction time in a heated **J-box** or roller steamer, where the fabric accumulates in folds so that line speed and dwell time are decoupled.

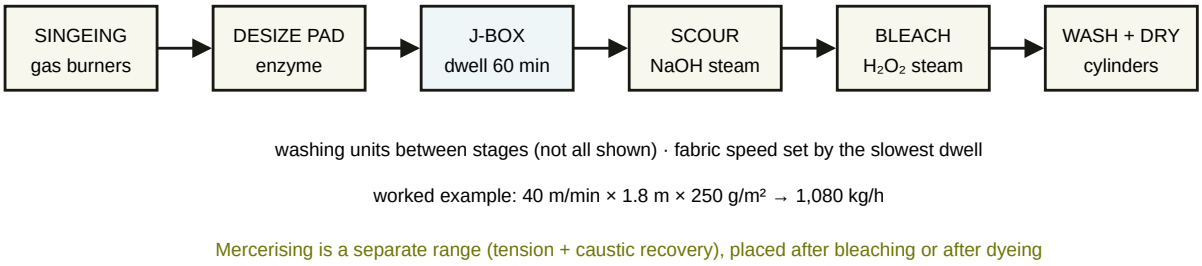


Fig. 1 — Continuous preparation range for woven cotton (schematic). Each stage is pad → dwell → wash; the range speed is fixed by the stage needing the longest dwell.

WORKED EXAMPLE 1 — RANGE THROUGHPUT

Range speed 40 m/min; fabric width 1.8 m; fabric mass 250 g/m² = 0.25 kg/m².
Mass per metre = 1.8 × 0.25 = 0.45 kg/m. Throughput = 40 m/min × 60 min/h × 0.45 kg/m = **1,080 kg/h**.
At this speed a 60-minute J-box dwell must hold 40 × 60 = 2,400 m of fabric — the physical size of the dwell unit is set by this product, which is why range speed cannot be raised without lengthening dwell capacity.

3. Batch preparation machines

Kier: a vertical pressure vessel in which cotton goods are boiled in caustic liquor circulated by pump or injector — economical and thorough, but hours-long and batchwise. **Soft-flow/jet machines** (Topic 7) prepare knits: gentle rope transport avoids the creasing a range would set into knitted structures. **Pad–batch** units apply desize or bleach liquor and hold the rolled fabric for hours at room temperature — low energy, suited to small lots. **Chain mercerisers** hold the fabric to width on pins/clips through the caustic and washing zones (best width control); **chainless** machines use roller paths and are preferred for some knits and heavy fabrics; both recover caustic soda by evaporation for reuse — the economics of mercerising turn on recovery.

4. What preparation quality is measured by

- **Absorbency:** drop test (seconds for a water drop to sink) — the scour's report card; uneven absorbency becomes uneven dyeing.
- **Residual size:** iodine test for starch; TEGEWA rating for film formers.
- **Whiteness index:** measured instrumentally after bleaching; also the base against which pale shades are predicted (Topic 1).
- **Damage:** fluidity (cuprammonium) or tensile loss — over-bleached or over-scoured fabric is weaker even when it looks perfect.
- **pH and residual alkali:** goods entering dyeing must be neutral-to-slightly-acid and uniformly so.

5. Glossary — key terms

- **Saturator:** the padder that impregnates fabric with process liquor at the head of a range stage.
- **J-box:** a heated accumulation chamber giving dwell time at continuous throughput.
- **Kier:** a pressure vessel for batch scouring/boiling of cotton goods.
- **Mercerising:** treatment of cotton with strong caustic soda under tension for lustre, strength and affinity.
- **Fluidity:** a measure of cellulose degradation (higher fluidity = more damage).

6. Key points

- Preparation is a fixed order of removals; each stage's machine is chosen by goods form (range for woven volume, batch/soft-flow for knits and lots).
- Continuous range speed is limited by dwell: throughput (kg/h) = speed × width × fabric mass per area — and dwell capacity scales with it.
- Absorbency, whiteness, residual size and damage are the four measurements that release fabric to dyeing.

7. Summary

Preparatory machinery is unglamorous and decisive. The range in Fig. 1 moving 1,080 kg/h is not merely fast — it must be uniformly fast in chemistry as well as motion, at every stage, because each later process in this course (dyeing, printing, finishing) will faithfully reproduce whatever preparation leaves uneven.

8. Practice

1. A range runs at 35 m/min with 1.6 m wide fabric of 200 g/m². Calculate throughput in kg/h, and the fabric content of a J-box giving 45 minutes dwell.
2. Why is desizing placed before scouring in the range sequence?
3. Give two reasons knitwear is prepared in soft-flow batch machines rather than on a woven preparation range.
4. What does caustic recovery contribute to mercerising economics?

Answer key

1. Mass per metre = 1.6 × 0.200 = 0.32 kg/m → throughput = 35 × 60 × 0.32 = 672 kg/h. J-box content = 35 × 45 = 1,575 m.
2. The size film waterproofs the fabric surface: scouring liquor cannot reach the fibre impurities uniformly through it, so scouring before desizing is patchy and wastes alkali. Removing the film first gives the scour access to the fibre.
3. Knits are crease- and distortion-sensitive: rope-form batch transport avoids setting creases and stretching that open-width tensioned ranges cause; knit lots are also typically shorter and more varied, suiting batch working.
4. Caustic soda is the merceriser's main consumable and a major effluent load if discharged. Evaporative recovery returns most of it to process strength, cutting chemical purchase and neutralisation cost — often the difference between mercerising being profitable or not.

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